AI And Machine Learning

How “Deep Industry Research Agents” Can Change Your Organization

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Summary. Corporate leaders often look to AI for breakthrough innovations, but a faster and more reliable payoff lies in using it to radically improve services productivity by eliminating low-value work. “Deep industry research agents”...

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Most corporate investments in AI today are bets on new business models, transformed customer experiences, entirely new products. Those may come. But there’s a much more immediate, concrete payoff that many leaders are overlooking: the use of AI to dramatically improve services productivity by eliminating wasted time.

Privacy - Terms

Services account for roughly 80% of U.S. GDP. The financial sector, in particular, has recently contributed more to GDP growth than any other sector. Yet services productivity remains poorly understood and poorly measured. In the absence of good measures for evaluating services productivity, most leaders fall back on what they can easily track: headcount. This, in turn, causes them to focus on cutting labor costs, which they do primarily by outsourcing or offshoring. But those practices are reaching their limits.

So what can you do if you want to improve your company’s productivity? You’ll need to focus less on *who* is doing the work and more on *what* work is being done.

We put that idea to the test recently in working with PIMCO, a global asset manager that oversees $2.26 trillion in assets. To analyze the work being done at the firm, we deployed AI-powered tools that we call *deep industry research agents*, or DIRAs. These agents were able to identify and significantly improve inefficiencies and demonstrated that it is indeed possible to achieve much higher productivity levels.

Below we’ll explain what DIRAs are, how we deployed them, and how they can help you boost the productivity of your services business by reducing wasted time, not headcount.

What Deep Industry Research Agents Do

Most executives are now familiar with the idea of AI agents—systems that, powered by an LLM and following a general set of instructions, can autonomously break down a task, draw on other tools or software, and feed their results back into a larger workflow.

Think of DIRAs as specialized AI analysts for different industries. They add two main elements to the mix. The first is *domain specificity*: They’re configured for a specific industry and set of operational problems. The second is *end-to-end case handling*: They can plan investigations, execute them using industry tools and data, and then make conclusions and recommend action.

In services organizations, most work takes the form of a “case.” In hospitals, clinicians work on patient cases; in customer service, agents work on customer cases; and in financial operations, teams work on exception cases. Cases vary enormously in time, cost, and business impact, but most management systems treat them all as if they’re interchangeable, which means that teams routinely spend too much time on minor issues and too little time on major ones. Traditional automation doesn’t solve this problem; in many instances, it simply accelerates the creation of low-value work.

Case Studies

In many financial institutions, nearly 70% of operations time is spent researching and resolving exception cases generated by reconciliation systems. So in the work we did with PIMCO, we decided to use DIRAs to diagnose and resolve a set of exception cases—discrepancies between investment and accounting records that are brought to the surface by automated checks, which in the typical flow of work would require human investigation to resolve. Critically, our DIRAs didn’t just try to solve these individual cases. For each one they also recorded key metrics such as time-to-resolve, financial impact, and root cause, and predictive accuracy (how well the agent’s initial assessment matched the final outcome).

Over more than eight months in daily production, our DIRAs processed tens of thousands of exception cases, which allowed PIMCO’s leaders for the first time to quantify exactly where operations teams were spending their time, and how much of that time generated real value. When we took stock of the results with

them, one insight stood out: A very large share of the exceptions were not true errors. Instead, they were explainable differences caused by known and legitimate variations in methodology. Discrepancies between investment and accounting records often arose, for example, because different teams were using different conventions for booking foreign exchange. For one set of funds, 76% of the root causes fell into this category. The apparent error, a difference in cash balances, turned out to be the predictable consequence of the way the systems and teams were configured.

Once the DIRAs learned these patterns, they were able to automatically recognize these known differences, reconcile them without requiring manual investigation, and flag only true anomalies for human review—a shift that, by eliminating low-value investigative work, could potentially free 65%–70% of the team’s time.

Deploying Your Agents

Our experience suggests that you can reap the benefits of DIRAs if you treat them as learning systems embedded in real operations, not as one-off technology projects. If you’re considering deploying them in your organization, we recommend taking six main steps.

1. Configure agents to learn from narrow, high-value use cases.

Executives are often tempted to launch “enterprise-wide” AI agents. That’s usually a mistake, because agents, like people, learn faster and perform better when their jobs are tightly defined. In configuring DIRAs, you should select a single, well-bounded type of case (for example, a specific reconciliation-break type or claims category); constrain the set of plausible root causes; and measure performance against clear, operationally relevant metrics. This focus improves learning efficiency, increases accuracy, and quickly generates tangible value, which, in turn, builds organizational confidence and support for scaling.

2. Put agents into real workflows.

Proof-of-concept pilots and test data have their place, but they’re not where real learning occurs. The operational environment is simply too complex and dynamic to be reproduced in a lab. Instead, after configuring your DIRAs you should insert them quickly into the daily workflow. Initially, keep them in an advisory mode, able to identify root causes and propose resolutions but not make final decisions. During a defined observation period (often several months), monitor their accuracy, identify their failure modes, and refine the models and rules. Only once you’ve determined that their performance is stable and well understood, you can allow auto-resolution, but even then do so only for low-risk, well-characterized case types, with clear escalation paths.

3. Use data gathered by agents to create a new management information system.

Because DIRAs generate a robust, standardized dataset about how work gets done, over time they can help you answer questions such as: Which case types consume disproportionate time relative to their business impact? Where are we repeatedly addressing symptoms rather than fixing root causes? Which processes or counterparties generate the most high-impact exceptions? This process is analogous to what ERP systems did for manufacturing and global supply chains: By making work and flows visible, they enabled new forms of performance management. DIRAs can play a similar role in services, using granular, case-level data to develop a new management-information system.

4. Prioritize agents that accumulate experience.

Many organizations start their AI journey by experimenting with general-purpose LLMs. These systems are powerful, but on their own they don’t accumulate firm-specific operational experience. For productivity-focused applications, leaders should insist that agents: continuously learn from resolved cases, not just static

training data; capture not only their own recommendations but the final human decisions and outcomes; and use that feedback to adjust future predictions, tuning themselves over time to the organization’s real-world patterns and preferences.

5. Extend agents across your operational ecosystem.

Most services organizations rely heavily on third-party providers and partners. Work and, therefore, exceptions, flow across organizational boundaries. If each party uses its own tools and data, resolution is slow and blame is diffuse. A better approach is to treat the agent platform as shared infrastructure.

PIMCO, for example, rolled out DIRAs with two of its major service providers. All three organizations now see the same cases, metrics, and identified root causes in real time. This shared view has reduced back-and-forth email and manual reconciliation, clarified the ownership of fixes when problems span firms, and created a common language for discussing performance.

6. Insist on auditability, security, and compliance from day one.

Regulatory, risk, and security concerns are a major and valid barrier to AI adoption in financial services and other regulated sectors. Many leaders understandably resist “black box” systems that no one can explain to auditors or supervisors. The response to that objection should be not to avoid agents but rather to set and enforce a “glass box” standard. Agents should produce a traceable audit trail for every case, showing what data they accessed, how they reasoned, and how confident they were. They should also integrate with existing controls, logging, and monitoring systems; operate within the enterprise’s security perimeter; and adhere to established data-governance rules. When agents are as transparent and controllable as other core systems, they can be scaled without compromising the firm’s risk posture.

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The dominant narrative about using agentic AI in services is that it will replace large numbers of workers. Our experience suggests instead that DIRAs will usher in a new era of productivity, by developing metrics that guide workers to spend the right time on the right tasks. In many financial-operations functions, for example, current systems generate huge volumes of false positives and low-value exceptions, and workers spend most of their time on those exceptions. DIRAs can help liberate them from this drudgery.

In the environment we studied, we found that the use of DIRAs can boost productivity by up to 70%. To put that figure in perspective, consider that long-term median total-factor productivity growth in services is roughly 1.7% per year. A 70% improvement would effectively compress about 31 years of average productivity growth into a single wave of change. Productivity gains of that nature could be a game changer for the services industry and the U.S. economy.

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